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Mathematics – L3 Applied Mathematics

**Family Name:**

**First Name:**

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Duration: 40 minutes

# Test in Information System and Databases

**Exercise 1: (05 points)**

Answer each of the following questions by checking the right box:

Q1- To share my printer with my coworkers, we need to be at least in the same:

☐ Office ☐ Intranet ☐ Internet ☐ Other

Q2- The Administrator of a database is:

☐ The Creator of the database ☐ An input device ☐ A regular user of the database ☐ Other

Q3- Matlab is:

☐ An input device ☐ A utility software ☐ A productivity software ☐ Other

Q4- Ubuntu is:

☐ An operating system ☐ A utility software ☐ an application software ☐ Other

Q5- A Laser printer is:

☐ An input device ☐ An output device ☐ an input-output device ☐ Other

**Exercise 2: (6 points)**

Given the following table **Customer** table in the relational database: **Store.** Answer the following questions:

Customer(customer_id, first_name, last_name, email)

| customer_id | first_name | last_name | email |
|---|---|---|---|
| 1 | JAMES | MILLER | james.miller@sakilacustomer.org |
| 2 | EMILY | DAVIS | emily.davis@sakilacustomer.org |
| 3 | MICHAEL | SMITH | michael.smith@sakilacustomer.org |
| 4 | SARAH | JOHNSON | sarah.johnson@sakilacustomer.org |
| 5 | DAVID | JONES | david.jones@sakilacustomer.org |
| 6 | JESSICA | BROWN | jessica.brown@sakilacustomer.org |
| 7 | ROBERT | WILSON | robert.wilson@sakilacustomer.org |

Q1- What is the: **(1 point)**

- degree of the table: , the number of records: ,the number of rows: , the number of attributes:

Q2- What is the data type of **(2 points)** :

- customier_id: , first_name:

Q3- What is the domain of values of **(2 points)**:

- customer_id:
- email:

Q4- We want that each customer can have only 2 phone numbers to contact him. What are the modifications to apply? **(1 point)**

**Exercise 3: (04 points)**

Fill in the blank:

My team and I gathered today to discuss the development of the application we are creating for our new customer: a book store owner. The project is small (not big), so we are not going to use the …… development method. We will not be using any of the Agile methodologies. Nevertheless, we will start by specifying all the requirements of that application. So, we will be using the …… development methodology. We will after organize …….. work sessions with a user's representative, for the design and the …..…... of the application.